Ken Yamashiro

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Education

University of Southern California
– B.S. Mechanical Engineering, GPA 3.7 — Expected Graduation May 2027
– Planning to pursue M.S. in Mechanical Engineering at USC — 2027 – 2028

Experience

USC Viterbi School of Engineering — Unmanned Aerial Systems Project — December 2025 – Present
Undergraduate Assistant and Lead Student Engineer — Los Angeles, CA

- Characterize flight dynamics of organic, bio-inspired aerodynamic designs on low-Reynolds-number autonomous VTOL drones in hybrid water/air applications under Dr. Penkova
- Develop electrical systems, program flight dynamics, and essential systems for VTOL flight

Niagara Bottling — May 2026 – August 2026
Packaging Research and Development Intern — Diamond Bar, CA

- Built multi-variable regression models using ANSYS FEA outputs to predict and validate bottle lightweighting designs, cutting simulation turnaround time by 1,000+ hours for the engineering team
- Integrated simulation outputs into 60+ design iterations to create predictive load models for production

NASA Columbia Scientific Balloon Facility — May 2025 – July 2025
Intern/Undergraduate Assistant — Palestine, TX

- Supported the final production and payload integration of the Payload for Ultrahigh Energy Observations (PUEO) research project, a NASA Pioneers project successfully launched in December 2025
- Performed surface preparation, resolved tolerance conformities, and prepared 350+ flight components
- Effectively collaborated with over 50 researchers from 11 universities across the U.S., UK, and Taiwan

UH Department of Physics and Astronomy, High Energy Physics Group (HEPG) — September 2023 – August 2025
Lab Tech II and Researcher — Honolulu, HI

- Designed, fabricated, and tested 8+ physics research projects focused on particle physics, radio frequency (RF), and space environments using Siemens Solid Edge, FEA, and drafting
- Fabricated 700+ flight hardware parts using CNC, CAM, manual milling, lathes, and additive manufacturing
- Prepared technical materials and assisted in proposal presentations to NASA, JPL, and various universities

Activities & Leadership

USC Formula SAE Racing | Powertrain Engineer — August 2025 - Present

- Designed, manufactured, and tested powertrain and cooling subsystem components for the SCR26 race car (600cc inline-4 Yamaha R6) using Siemens NX, ANSYS, and MATLAB
- Selected from 60+ members to present in the design competition at the 2026 Michigan FSAE competition
- Helped place 7th out of 110 teams by supporting assembly, physical testing, and analysis of the SCR26 car

USC Viterbi School of Engineering | Mechoptronics Lab Course Producer — August 2026 - Present

- Support instruction for a 200-student Mechoptronics course by running weekly lab sessions and assisting students with circuit assembly, actuator control, and troubleshooting.
- Grade lab reports, providing technical feedback on methodology and documentation

Project Experience (More listed on GitHub Website)

Cosmic Ray Lunar Sounder (CoRaLS) — NASA JPL/UH HEPG

- Fabricated and tested 8+ proprietary antenna prototypes for a lunar-ice-detection spacecraft
- Contributed to the initial mechanical design of a deployable antenna assembly that stows and unfolds to full aperture in orbit, balancing stowed volume, deployment reliability, and RF geometry

Askaryan Calorimeter Experiment (ACE) — Fermilab/UH HEPG

- Fabricated and bench-tested RF waveguide prototypes used to validate a new coaxial-to-waveguide adapter design, resolving tolerances and manufacturability for a geometry unsuited to standard machining

Skills and Certifications

Skills: SolidWorks, Siemens NX, ANSYS FEA, CFD, MiniTab, Manual/CNC Machining, GD&T, 3D printing, MATLAB
Experience in: Additive/Subtractive manufacturing, 6061, 7075 Al, Brass 360, CF tubes/honeycomb, 100/101 Cu, more
SOLIDWORKS CAD Design Associate Certification, Certificate ID: C-SY4HKFETCM